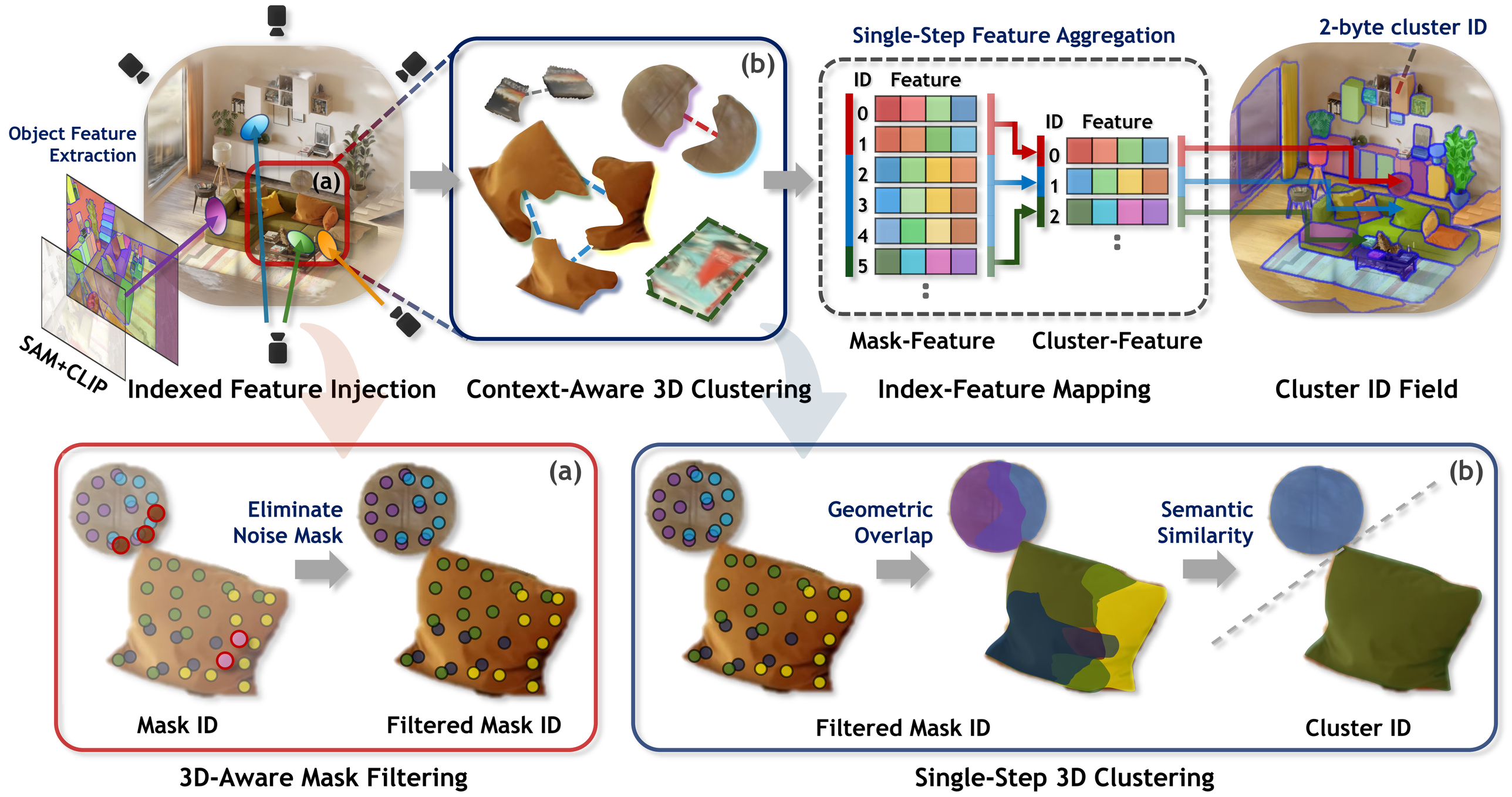
Object Feature Extraction
SAM+CLIP
(a)
Indexed Feature Injection
(b)
Context-Aware 3D Clustering
Single-Step Feature Aggregation
ID
Feature
0
1
2
3
4
5
ID
Feature
0
1
2
Mask-Feature
Cluster-Feature
Index-Feature Mapping
2-byte cluster ID
Cluster ID Field
(a)
Eliminate Noise Mask
Mask ID
Filtered Mask ID
3D-Aware Mask Filtering
(b)
Geometric Overlap
Semantic Similarity
Filtered Mask ID
Cluster ID
Single-Step 3D Clustering